SHOOTERSAM · MAP1 · UNREAL ENGINE 5.7

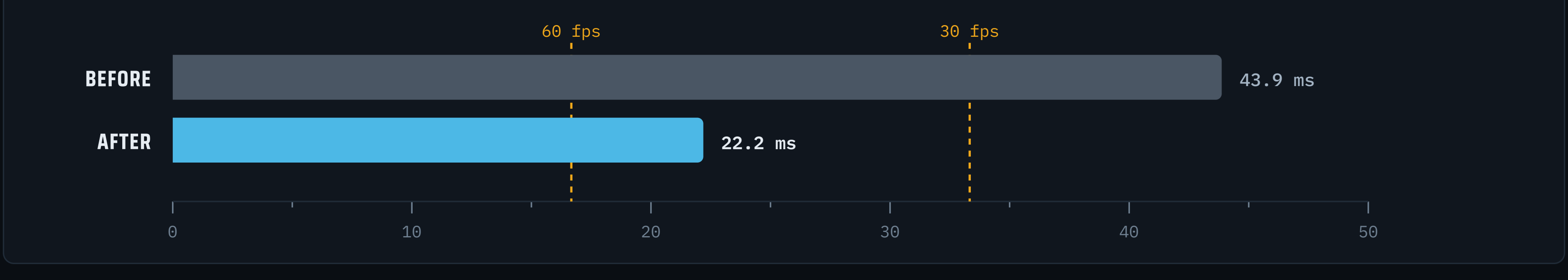
LUMEN ON A MACBOOK AT HALF THE COST

I took a sci-fi interior map built for high-end lighting and made it run well on an M3 Pro MacBook, with four quality tiers: baked lighting on Low and Medium, software Lumen on High and Epic. Every number on this page comes from the same 30-second camera flight in a packaged build.

`Apple M3 Pro` `macOS 26.2 · Metal` `Packaged Development build` `1512×982` `120 Hz display`

EPIC · GPU TIME PER FRAME

lower is better · ms



−49%
GPU time on Epic, 43.9 to 22.2 ms per frame

37 → 64.5
fps on High, with Lumen on

22 → 43.5
fps on Epic

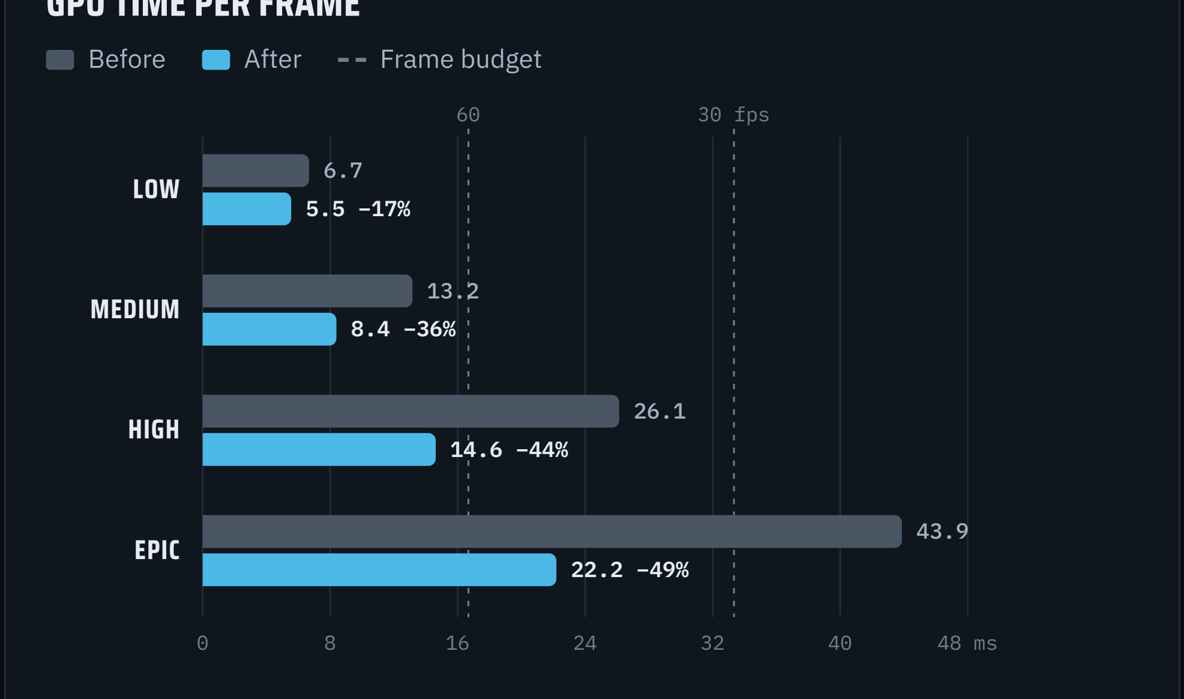
0
hitches during the Epic flight

BEFORE AND AFTER

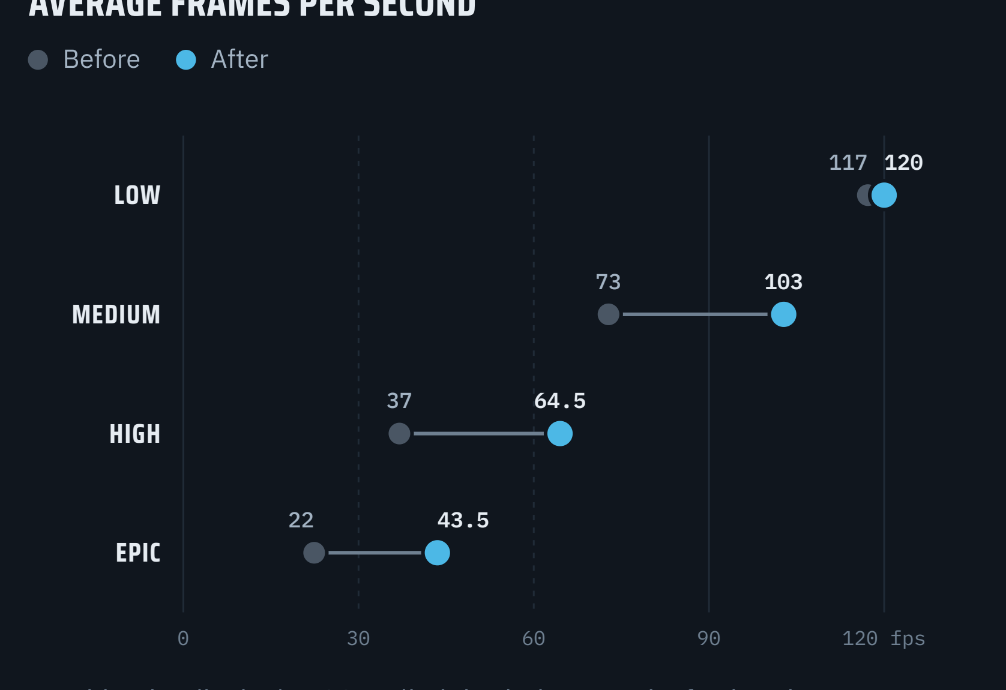
EVERY TIER GOT FASTER

Before: the original map and assets in a clean project, each tier set with Unreal's own scalability presets. After: the optimized map with its tier settings. Both at the same render resolution per tier, on the same flight.

GPU TIME PER FRAME



AVERAGE FRAMES PER SECOND



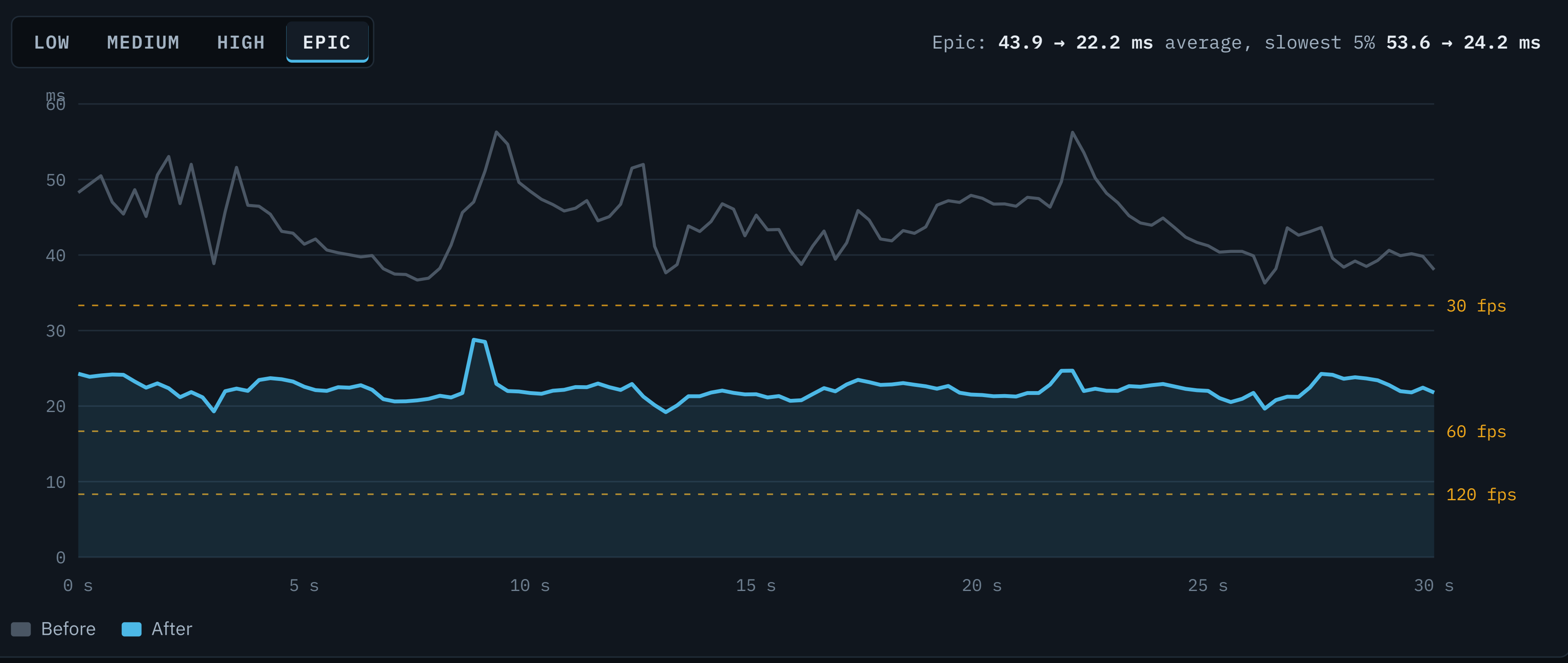
Low hits the display's 120 Hz limit both times, so its fps barely moves even though its GPU time dropped 17%.

THE FLIGHT

FRAME BY FRAME

GPU time across the 30-second flight, before and after, averaged into quarter-second steps. The dashed lines are the budgets for 120, 60 and 30 fps.

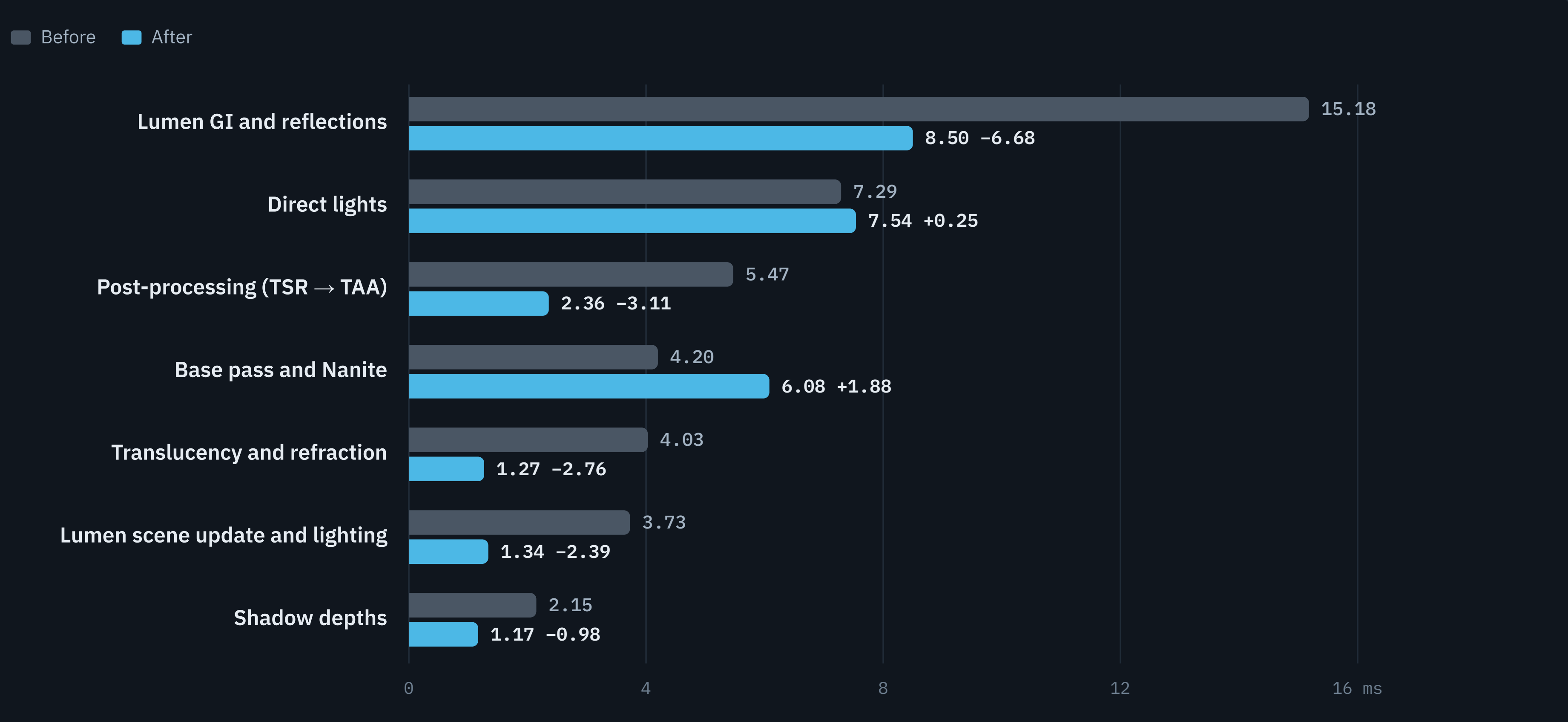
Epic: **43.9 → 22.2 ms** average, slowest 5% **53.6 → 24.2 ms**



WHERE THE TIME WENT

ONE FRAME, PASS BY PASS

A single Epic frame from the same corridor, captured with ProfileGPU in both projects at about the same render size (1.3 megapixels). The scene went from 48.5 to 30.1 ms.



Direct lighting stayed about the same: shadowed lights were cut, but more unshadowed fill lights were added for the look. Base pass rose slightly with Nanite; the instancing win is on the CPU, in draw calls. Times from one editor frame, so expect some noise.

THE TIERS

WHAT EACH TIER DOES

A C++ quality manager in the map applies each tier at runtime: Unreal's scalability level, the tier's console variables, its post-process volume and its own set of lights. Nothing is written to project config, and everything is restored when play ends.

LOW

Scalability 0

BAKED LIGHTING

FPS	GPU MS
117 → **120**	6.7 → **5.5**

RENDER	60% · 907×589, TAA
GI	Baked lightmaps
REFLECTIONS	Reflection captures
LIGHTS	Low set, baked shadows
LOOK	PPV_Low

`r.ScreenPercentage 60`
`r.Lumen.DiffuseIndirect.Allow 0`
`r.Lumen.Reflections.Allow 0`

MEDIUM

Scalability 1

BAKED LIGHTING

FPS	GPU MS
73 → **103**	13.2 → **8.4**

RENDER	70% · 1058×687, TAA
GI	Baked lightmaps
REFLECTIONS	Screen space
LIGHTS	Low set, baked shadows
LOOK	PPV_Epic, Lumen off

`r.ScreenPercentage 70`
`r.Lumen.DiffuseIndirect.Allow 0`
`r.Lumen.Reflections.Allow 0`

HIGH

Scalability 2

SOFTWARE LUMEN

FPS	GPU MS
37 → **64.5**	26.1 → **14.6**

RENDER	80% · 1210×786, TAA
GI	Lumen, scene updates spread over 64 and 128 frames
REFLECTIONS	Lumen, traced up to roughness 0.25
LIGHTS	High set, every light shadowed
LOOK	PPV_Epic

`r.ScreenPercentage 80`
`r.Lumen.Reflections.MaxRoughnessToTrace 0.25`
`r.LumenScene.DirectLighting.UpdateFactor 64`
`r.LumenScene.Radiosity.UpdateFactor 128`
`r.Lumen.TraceMeshSDFs 0`

EPIC

Scalability 3

SOFTWARE LUMEN

FPS	GPU MS
22 → **43.5**	43.9 → **22.2**

RENDER	90% · 1361×884, TAA
GI	Lumen with short-range AO
REFLECTIONS	Lumen, traced up to roughness 0.4
LIGHTS	Epic set, key lights shadowed
LOOK	PPV_Epic

`r.ScreenPercentage 90`
`r.Lumen.Reflections.MaxRoughnessToTrace 0.4`
`r.Lumen.ScreenProbeGather.ShortRangeAO 1`
`r.LumenScene.DirectLighting.UpdateFactor 64`
`r.LumenScene.Radiosity.UpdateFactor 128`
`r.Lumen.TraceMeshSDFs 0`

THE WORK

WHAT I CHANGED

Each figure is tagged with where it came from. *One frame* is the same-corridor ProfileGPU comparison above; *editor* and *standalone* are working measurements taken along the way, so they show the size of a change rather than final numbers.

STATIC MESHES TO ISM + NANITE

I wrote an editor tool that converts placed static meshes into instanced static meshes (ISM) and rebuilt the level's structure as instanced Blueprints from one base Blueprint, with Nanite on the meshes. Instancing cuts draw calls; Nanite handles the geometry detail.

~2,500 → **~650 draws** `STAT UNIT · SEPARATE SESSIONS`

DRAW AND CULL DISTANCES

Max draw distances on small props, decals and lights, with fade distances on lights so they don't pop. Cull distance volumes handle the rest of the props by size.

FEWER SHADOWS, TIGHTER LIGHTS

15 ms → **4.4 ms** `EPIC · EDITOR`

Dynamic shadows only where they read, across the whole scene. Smaller attenuation radii and tuned falloff (inverse-square setting and falloff exponent) mean each light touches fewer pixels. In the same-frame capture, shadow depths went from 2.2 to 1.2 ms. Each shadow-casting light renders its own shadow pass and is lit into Lumen's scene; cutting them also took Lumen's scene lighting from 2.9 to 0.4 ms.

CHEAPER MATERIALS

4.0 ms → **1.3 ms** `EPIC · ONE FRAME`

Refraction and mirror-like reflective materials kept to the few places that need them, which removed the refraction pass (1.6 ms) and cut the translucency lighting volume from 1.4 to 0.3 ms. Two Sided is off on every master material.

HARDWARE TO SOFTWARE LUMEN

18.9 ms → **9.8 ms** `EPIC · ONE FRAME`

Software Lumen instead of hardware ray tracing, with scene lighting updates spread over 64 and 128 frames, coarse distance-field tracing, and traced reflections only on shiny surfaces.

TAA INSTEAD OF TSR

3.8 ms → **0.96 ms** `EPIC · ONE FRAME`

TSR's cost follows the output resolution and was too high for this GPU. Each tier renders at its own share of a fixed window and TAA upscales it. Post-processing overall went from 5.5 to 2.4 ms.

TEXTURE MEMORY

1.42 GB → **1.18 GB** `EDITOR, INCLUDES EDITOR MEMORY`

Virtual textures for the materials and virtual lightmaps for the bake, so only the visible parts stay in memory. Then I used the level's Size Map to find the heaviest textures and fixed their compression.

BAKED LIGHTING ON APPLE SILICON

Low and Medium use baked lightmaps. UnrealLightmass runs under Rosetta on Apple Silicon and crashed in Embree; turning Embree off made baking work.

SUN SHAFTS, KEPT OUTSIDE

The directional light stays on for its light shafts. Invisible shadow-casting blockers keep it out of the interior.

A MAC-ONLY STALL

50–100 ms → **none** `EPIC · STANDALONE`

Mouse-look in fullscreen stalled the frame. Unreal Insights and Xcode Instruments traced it to `FMacApplication::OnWindowDidResize` forcing a render flush. Windowed mode avoids it.

FIRST-SHOT STUTTER

The muzzle flash and impact effects recompiled on first use every session. Impacts now come from a component pool and both effects are pre-warmed at load.

A REPEATABLE BENCHMARK

The pause menu switches tiers and runs a camera-rail flight between Unreal's `StartFPSChart` and `StopFPSChart`, so every run is the same path.

Method: each result is one 30.1-second run of the same Level Sequence camera flight in a packaged Mac Development build, 1512×982 windowed fullscreen with VSync on. GPU time and fps come from Unreal's FPS chart logs. The before runs used the original assets in a clean project with no player, enemies or music on screen, which if anything favours them.